

SHANGHAI HONGQIAO INTL, China

WMO: 583670

Lat: 31.198N

Lon: 121.336E

Elev: 10

StdP: 14.69

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.5	30.0	10.0	9.2	37.5	14.2	11.3	37.9	22.3	37.9	20.5	38.9	5.5	310	0.462

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	97.1	81.0	94.8	80.7	92.8	80.3	83.6	91.4	82.3	89.9	81.3	88.7	8.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.1	167.0	87.5	80.6	159.0	86.5	79.1	151.2	85.9	47.4	91.5	46.1	91.2	45.1	89.2	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.4	18.1	16.2	DB	23.8	100.8	2.8	2.3	21.9	102.5	20.2	103.8	18.7	105.1	16.7	106.8	
			WB	21.2	85.0	2.9	1.4	19.1	86.1	17.4	86.9	15.8	87.7	13.6	88.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	41.2	44.2	51.7	61.7	71.0	77.1	85.4	84.8	77.3	67.9	57.1	45.7	(6)	
(7)		DBStd	16.22	6.16	6.72	6.84	6.36	4.93	4.80	4.93	4.29	4.98	5.22	6.92	6.65	(7)	
(8)		HDD50	714	283	186	60	3	0	0	0	0	0	17	165		(8)	
(9)		HDD65	2760	739	583	415	137	8	0	0	0	30	249	599		(9)	
(10)		CDD50	5770	9	23	112	352	650	813	1096	1078	820	555	231	31	(10)	
(11)		CDD65	2343	0	0	3	37	193	363	631	613	370	120	13	0	(11)	
(12)		CDH74	23244	0	2	24	237	1129	2736	8066	7539	3037	445	29	0	(12)	
(13)		CDH80	9913	0	0	3	54	302	925	4130	3543	896	59	1	0	(13)	
(14)	Wind	WSAvg	8.3	8.0	8.4	8.9	8.8	8.8	7.8	8.8	9.1	8.4	7.6	7.6	7.7	(14)	
(15)	Precipitation	PrecAvg	50.8	2.8	2.8	3.7	3.4	3.9	8.6	5.7	8.2	4.5	2.8	2.5	2.0	(15)	
(16)		PrecMax	67.6	8.1	6.3	9.6	6.3	8.5	28.3	11.8	18.3	10.3	11.7	5.2	4.8	(16)	
(17)		PrecMin	34.3	0.6	0.7	1.1	1.5	0.8	1.1	1.1	0.7	0.6	0.2	0.4	0.0	(17)	
(18)		PrecStd	8.9	1.5	1.6	2.0	1.3	1.9	5.3	3.1	3.8	2.4	2.6	1.6	1.3	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.0	71.2	77.1	86.1	91.2	95.4	102.0	100.5	93.5	85.7	77.4	66.1	(19)	
MCWB			55.8	59.5	61.4	68.4	70.8	80.5	80.9	81.6	79.7	73.1	66.0	57.9	(20)		
2%		DB	57.6	62.6	71.4	80.4	86.2	91.6	98.5	96.7	90.0	80.9	73.5	62.3	(21)		
		MCWB	51.3	54.3	58.1	66.6	69.2	78.6	81.1	81.2	78.2	69.5	63.7	55.3	(22)		
5%		DB	53.8	57.5	66.5	75.5	82.8	88.1	95.4	93.5	87.6	78.7	70.2	58.9	(23)		
		MCWB	48.4	50.8	55.9	63.7	68.0	76.4	80.8	80.7	76.9	67.9	62.7	52.6	(24)		
10%		DB	50.3	53.9	62.6	71.9	80.5	85.6	93.3	91.5	84.5	76.6	67.8	55.7	(25)		
		MCWB	45.8	47.6	54.9	61.4	66.9	74.9	80.8	80.1	74.8	66.8	60.6	49.9	(26)		
(27)		Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.6	61.1	64.6	72.8	77.1	82.4	85.3	85.4	84.0	77.2	70.5	61.9	(27)
MCD B				61.9	68.4	70.7	80.2	85.9	91.1	93.3	92.8	88.9	82.2	74.1	64.7	(28)	
2%	WB		53.2	55.9	60.9	69.5	74.2	80.6	83.6	83.6	80.7	73.4	67.3	56.8	(29)		
	MCD B		56.5	60.5	67.3	75.9	80.6	88.8	92.1	91.1	86.0	77.8	71.0	60.3	(30)		
5%	WB		49.8	52.4	58.6	66.8	72.2	79.2	82.5	82.3	79.1	71.0	64.4	54.0	(31)		
	MCD B		52.1	57.2	64.6	72.9	78.0	85.6	90.8	89.1	84.6	75.3	68.2	57.6	(32)		
10%	WB		46.5	49.4	55.5	64.0	70.3	77.6	81.5	81.2	77.6	69.1	61.9	50.9	(33)		
	MCD B		49.5	52.7	61.4	70.3	75.6	82.9	89.6	88.0	83.2	73.8	66.2	54.7	(34)		
(35)	Mean Daily Temperature Range		5% DB	MDBR	11.8	12.5	13.9	15.0	14.4	11.7	12.4	11.3	11.7	12.9	13.3	13.0	(35)
MCD BR				16.8	19.0	20.7	20.7	19.2	16.5	15.5	13.9	13.3	14.8	15.6	16.5	(36)	
MCWBR		10.8		11.3	11.1	10.6	8.1	6.5	5.4	5.0	5.3	6.6	8.0	10.3	(37)		
5% WB		MCD BR	13.3	16.8	18.0	17.4	14.6	14.1	14.1	12.9	11.8	11.5	11.4	12.1	(38)		
		MCWBR	10.5	11.6	11.9	11.0	8.5	6.7	5.9	5.5	5.7	6.9	7.9	9.9	(39)		
																	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.598	0.638	0.671	0.679	0.663	0.717	0.685	0.660	0.614	0.612	0.620	0.590	(40)		
taud		1.664	1.611	1.574	1.616	1.681	1.619	1.715	1.750	1.816	1.739	1.677	1.692	(41)			
Ebn at Noon		188	197	204	211	216	204	210	213	216	201	181	181	(42)			
Edh at Noon		65	75	83	83	78	83	75	72	65	65	64	61	(43)			
(44)	All-Sky Solar Radiation	RadAvg	715	836	1184	1396	1503	1318	1673	1546	1255	1073	791	704	(44)		
(45)		RadStd	93	162	97	159	164	193	199	205	125	143	96	100	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) +0.68	(e) +0.83	(f) N/A	(g) +1.19	(h) N/A	(i) N/A	(j) N/A	(k) -196	(l) +241	(m) +135	(46)
(47)	Regional (10 neighbors)	+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+155	N/A	(47)

Nomenclature: See separate page